

Statistical Mechanics I Exercise 4

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1 Landau Theory For The Tri-critical Point

In this section we consider the following model that has a tri-critical point:

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - K \sum_{\square} \sigma_i \sigma_j \sigma_k \sigma_l - h \sigma_i \quad (1)$$

1.1 The phase diagram

To draw the phase diagram, let us notice a few things about the model:

- For $K = 0$, the model is reduced to the Ising model, meaning that at that point we get the well known second order phase transition.
- K is an interaction term, and so will have a similar effect on the critical temperature to that of J . So, $T^*(K)$ has a positive slope.
- Then, of course, the area $T < T^*(K)$ is the ordered phase with multiple solutions, while $T > T^*(K)$ is the unordered phase with a global minimum at $m = 0$.
- Since higher temperature pushes the system towards the center to $m = 0$, the line that separates the 3 solutions area from the five solutions area $K^*(T)$ has positive slope as well: the higher the temperature, the higher K^* needs to be to pass create the two additional solutions.
- Finally, the tricritical point is at the meeting of these two lines.

The phase diagram in the $K - T$ space can be seen in figure 1.

1.2 The Landau free energy

The Landau free energy can be seen in figure 2 at different parts of the phase space. As can be seen, at some parts of the phase space the free energy has five extrema. This is possible only for a polynomial with degree six or above. Since for $h = 0$ the model retains the $m \rightarrow -m$ symmetry, the polynomial has only even powers except for the linear term with prefactor h , leading to the final form of the free energy:

$$\Phi = -hm + am^2 + bm^4 + dm^6 \quad (2)$$

1.3 Critical exponents

Let us calculate the critical exponents α and β near the tricritical point. At that point, For these two exponents, we set $h = 0$ and derive to find the minimising magnetization:

$$\frac{\partial \Phi}{\partial m} = 2am + 4bm^3 + 6dm^5 = 0 \quad (3)$$

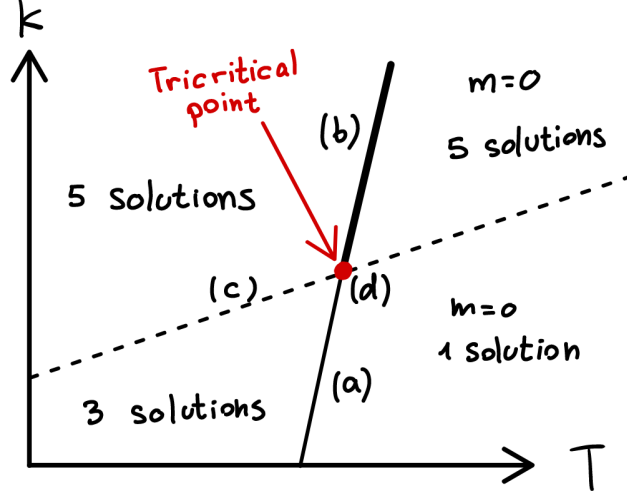


Figure 1: The phase diagram of the model in the $K - T$ space. (a) The second order phase transition line, (b) the first order phase transition line, (c) the line separating the 3-solutions and 5-solutions areas (d) the tricritical point

$$m = 0, m^2 = \frac{-b \pm \sqrt{b^2 - 3ad}}{3d} \quad (4)$$

At the tricritical point, all solutions converge, which requires $a = b = 0$. So, going along the line $d = \text{const}$ we can expand a and b as a function of $t = \frac{T-T_C}{T_C}$ and $k = \frac{K-K_C}{K_C}$ and keep only the first order. We care about the relation to t , so for small t , the smaller power is more significant, which is in this case the power of a which is $1/4$, giving $\beta = 1/4$:

$$m \sim \sqrt{\frac{-b \pm \sqrt{b^2 - 3ad}}{3d}} \sim \sqrt{\frac{-b_T t \pm \sqrt{b_T^2 t^2 - 3da_T t}}{3d}} \sim t^{1/4} \quad (5)$$

Now let us find α - the critical exponent of the heat capacity. For this, let us remember that the heat capacity can be calculated from the free energy using:

$$C = -\frac{\partial^2 F}{\partial T^2} \quad (6)$$

The free energy is

$$F = V\Phi(m_0, h = 0) = V(am^2 + bm^4 + dm^6) \quad (7)$$

Since $a, b \sim t$ and $m \sim t^{1/4}$, we get:

$$F \sim (t^{3/2} + t^2 + t^{3/2}) \sim t^{3/2} \quad (8)$$

$$C \sim t^{-1/2} \quad (9)$$

So we got $\alpha = 1/2$. Unsurprisingly, we got different critical exponents from the Ising model.

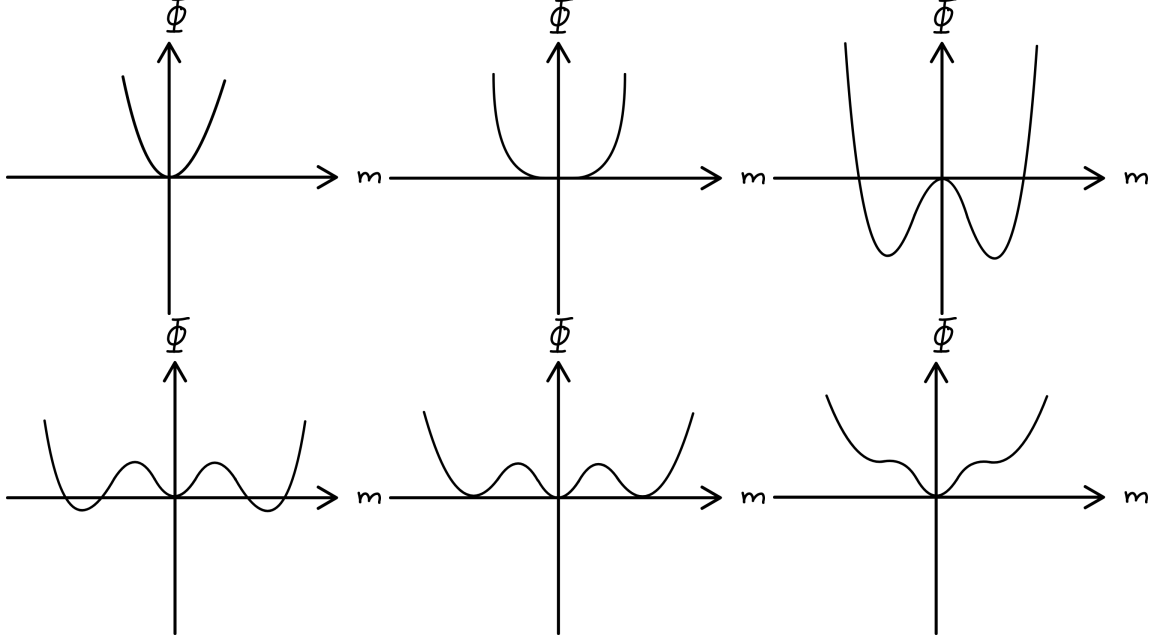


Figure 2: The Landau free energy at different parts of the phase space. Top row: (left) unordered phase with one minimum at $m = 0$ (middle) at the second order phase transition point (right) ordered phase with two minima at $m \neq 0$. Bottom row: (left) ordered phase with two minima at $m \neq 0$ (middle) at the first order phase transition point (right) at the line separating the 3-solutions and 5-solutions areas.

2 Landau Theory for the XY model

Let us consider the XY model:

$$H = \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j - \vec{B} \sum i \vec{S}_i \quad (10)$$

where $S_i = (\cos \theta_i, \sin \theta_i)$ for some real θ_i .

2.1 The Landau Free Energy

In this model, we can define an order parameter $\vec{S}$ which is the average spin configuration over an area. For an external field $B = 0$, there is a rotation symmetry $O(2)$, which means only the size of $\vec{S}$ matters, i.e. we get only even terms in $\vec{S}$. When $B \neq 0$, we get an additional $\vec{B} \cdot \vec{S}$ linear term:

$$\Phi(\vec{S}, T, B) = -\vec{B} \cdot \vec{S} + aS^2 + bS^4 \quad (11)$$

2.2 Critical Exponents

Let us now calculate the critical exponents α, β, γ , and δ . At $B = 0$:

$$\Phi(\vec{S}, T, B) = aS^2 + bS^4 \quad (12)$$

This is very similar to the Ising model with S instead of m , and so we expect the exponents to be the same. So, looking at $b > 0$, since that is where the system has minima, we get:

$$S_0^2 = \begin{cases} \pm \frac{a}{2b} & a < 0 \\ 0 & a > 0 \end{cases} \quad (13)$$

Which means a acts like $t = \frac{T-T_c}{T_c}$. On the other hand, b must be greater than zero, and so it has a zeroth order. So, around $t = 0 \rightarrow a = 0$, in leading order:

$$a \approx a_0(T - T_c) \quad b \approx b_0 \quad (14)$$

This gives:

$$S_0 = \sqrt{\frac{2a}{b}} \approx \sqrt{\frac{a_0 T_c t}{2b_0}} \sim t^{1/2} \quad (15)$$

So the first critical exponent is $\beta = 1/2$. Next, let us look at the susceptibility χ at $T > T_c$. For this, we go back to $B \neq 0$, and keeping again the first two terms, we can set $b = 0$. In this case:

$$\bar{S}_0 = \frac{1}{2a} \bar{B} \approx \frac{1}{2a_0 T_c t} \bar{B} \quad (16)$$

And so

$$\chi = \nabla_B \bar{S} = \frac{1}{2a_0 T_c t} \sim t^{-1} \quad (17)$$

Which gives, as expected $\gamma = 1$. Next let us calculate α . The free energy is given by

$$F = V\Phi(S_0, B = 0) = V \left(\pm \frac{2a^2}{b} + \frac{4a^2}{b} \right) \sim V \frac{a^2}{b} \sim t^2 \quad (18)$$

So

$$C \sim \text{const} \quad (19)$$

And we got the expected $\alpha = 0$. Finally, let us look at δ , which encodes the ratio of $\bar{S}$ and $\bar{B}$ at the critical point, rather than above it like γ . For this, let us take the model at $t = 0$ and $B \neq 0$. This means, since $a \sim t = 0$, we get:

$$\Phi \sim -\bar{B} \cdot \bar{S} + b_0 S^4 = -\bar{B} \cdot \bar{S} + b_0 S^2 (\bar{S} \cdot \bar{S}) = (-\bar{B} + b_0 S^2 \bar{S}) \cdot \bar{S} \quad (20)$$

$$\nabla_S \Phi = -\bar{B} + b_0 S^2 \bar{S} = 0 \quad (21)$$

$$\bar{S}_0 = \left(\frac{|\bar{B}|}{b_0} \right)^{1/3} \hat{B} \quad (22)$$

Which gives $\delta = 1/3$.

2.3 The susceptibility prefactor ratio

Let us look at the prefactors of the susceptibility above and below the critical point. We calculated χ_+ in the previous section, let us now calculate χ_- for $t < 0$. This time, since it takes part of the minimization, we cannot ignore b . For this, let us take S_0 to be the minimum $S_0^2 = \pm \frac{2a}{m}$ and expand around it $\bar{S} = \bar{S}_0 + \delta \bar{S}$ where $\delta \bar{S}$ encodes the effects of $\bar{B}$. This gives

$$\Phi = -\bar{B} \cdot \delta \bar{S} + a \delta S^2 + 2a \bar{S}_0 \cdot \delta \bar{S} + b \delta S^4 + 4b \delta S^2 \delta \bar{S} \cdot \bar{S}_0 + 2b \delta S^2 S_0^2 + 4b (\bar{S}_0 \cdot \delta \bar{S})^2 + 4b S_0^2 \bar{S}_0 \cdot \delta \bar{S} + f(S_0) \quad (23)$$

Where $f(\bar{S}_0)$ encodes all the information about $\bar{S}_0$ that we do not care about, since we are about to derive by $\delta \bar{S}$. Grouping the terms by order in $\delta \bar{S}$:

$$\Phi = b \delta S^4 + 4b \delta S^2 \delta \bar{S} \cdot \bar{S}_0 + (a + 2b S_0^2) \delta S^2 + 4b (\bar{S}_0 \cdot \delta \bar{S})^2 - \bar{B} \cdot \delta \bar{S} + (2a \bar{S}_0 + 4b S_0^2 \bar{S}_0) \cdot \delta \bar{S} + f(S_0) \quad (24)$$

Since by the definition of $\bar{S}_0$

$$2a \bar{S}_0 + 4b S_0^2 \bar{S}_0 = 2(a + 2b S_0^2) \bar{S}_0 = 0 \rightarrow a + 2b S_0^2 = 0 \quad (25)$$

so the first square term and the second linear term vanish. Keeping only up to second order in $\delta\bar{S}$, we get:

$$\Phi = 4b (\bar{S}_0 \cdot \delta\bar{S})^2 - \vec{B} \cdot \delta\bar{S} + f(S_0) \quad (26)$$

Deriving by $\delta\bar{S}$ we get:

$$\nabla_{\delta S} \Phi = 8b (\bar{S}_0 \cdot \delta\bar{S}) \bar{S}_0 - \vec{B} = 0 \quad (27)$$

Which is solved by:

$$\delta\bar{S} = \frac{1}{8bS_0^2} \vec{B} = \frac{1}{4a_0} \vec{B} \quad (28)$$

And so we get $A_+ = \frac{1}{2a_0}$ and $A_- = \frac{1}{4a_0}$ so the ratio $A_+/A_- = 2$ matches the ratio we found for the Ising model.